MIL-F-3967D
20 November 1980
SUPERSEDING
MIL-F-3967C
7 January 1975

MILITARY SPECIFICATION

FORMING MACHINES, SHEET METAL, ROTARY, COMBINATION, FLOOR MOUNTING, MOTOR DRIVEN

This specfication has been approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers floor mounting, electric motor driven, rotary, combination forming machines for performing beading, burring, crimping, flanging, slitting, turning and wiring operations on sheet metal.

1.2 Classification. The forming machines covered by this specification shall be of the following sizes. The size supplied shall be as specified (see 6.2.1).

Size 6	6 inch - 152mm throat depth
Size 12	12 inch - 304mm throat depth
Size 20	20 inch - 508mm throat depth

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3441

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:04Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services

MILITARY

MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and the publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

C113.1 - Electrical Standards for Metalworking Machine Tools (NFPA No. 79)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York 10018).

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish a sample unit for first article inspection and approval (see 4.2, 6.2.1 and 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage or distortion shall be accessible for adjustment, replacement, and repair.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:04Z
Check the source to verify that this is the current version before use.

3.2.1 Reclaimed materials. The machine shall contain reclaimed materials to the maximum extent possible without jeopardizing its intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which restores them to the same chemical composition and physical properties as the materials originally selected for use on the machine. Reclaimed materials shall include iron, steel, copper, brass, aluminum, fiber products, plastics, and elastomers that have been collected from discarded solid, liquid, semi-solid, or gaseous waste such as garbage, refuse, or sludge.

3.2.2 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.3 Safety and health requirements. Covers, guards, and other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair. All machine parts, components, mechanisms, and assemblies, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine. Additional safety and health requirements such as those required for integration into the operating environment, shall be as specified (see 6.2.1 and 6.4).

3.2.4 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Each recirculating system shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.5 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner that prevents change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

3.3.3 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:04Z
Check the source to verify that this is the current version before use.

All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.5 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.6 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.3.7 Gears. All gears shall be constructed of a material suitable for their application. Gears shall be applied in a manner to preclude either premature damage, wear or failure.

3.4 Components. The machine shall consist essentially of a pedestal or base, frame, rolls and their shafts, roll closing and adjusting mechanism.

3.4.1 Pedestal or base. The pedestal or base shall be of sufficient size and have the capacity to rigidly support the mounted components while the machine is operating at its maximum capacity. Not less than three through holes shall provide for mounting the pedestal or base to the floor or a foundation.

3.4.2 Frame. The frame shall mount to the pedestal or base and have provisions to rigidly support the motor, drive mechanism, roll shafts and rolls.

3.4.3 Rolls and roll shafts. The rolls and roll shafts shall be constructed of a material suitable for the application. The roll shafts shall be supported by bearings in a mounting suitable to their application, which includes sleeve type bearings.

3.4.4 Adjusting and closing mechanism. An adjusting and closing mechanism shall provide manual adjustments to include the complete range for standard rolls furnished with the machine. Means shall be provided to laterally adjust one roll for maintaining alignment of the rolls. A pneumatic cylinder may be provided to open and close the rolls provided all necessary hoses, piping valves, regulators and controls are furnished to connect into a 90 to 120 psi shop air supply.

3.4.5 Clutch. Machines having a clutch shall provide engagement and release of the clutch, with the motor running, upon actuation of manual controls.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:04Z
Check the source to verify that this is the current version before use.

3.4.6 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to all requirements in this paragraph. The system shall conform to the requirements of ANSI C113.1. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit having a fused safety disconnect switch. The machine shall be initially wired for operation on 230 volts. An identified terminal for grounding the machine when installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in Table 240-A of ANSI C113.1 for the disconnect fuse rating.

3.4.6.1 Motors. Motors shall have ball or roller bearings. Unless otherwise specified (see 6.2.1), each motor shall meet the requirements of a dripproof enclosure.

3.5 Machine characteristics. The machine shall meet the requirements of Table I.

TABLE I

Sizes and Characteristics.

Characteristics	Size 6	Size 12	Size 20
Throat Depth, Inches	6	12	20
Motor horsepower	1/4	1	2
Roll Speed (feet per minute)	17	15	21
Workpiece capacity:			
Slitting and Circle Cutting	16 ga	14 ga	10 ga
Crimping	20 ga	16 ga	10 ga
Combined Crimping & Beading	22 ga	18 ga	12 ga
Other Standard Operations	18 ga	14 ga	10 ga

NOTE: Requirements are not less than those shown.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:04Z
Check the source to verify that this is the current version before use.

3.6 Performance. The machines covered by this specification shall be capable of performing the operations listed in table I on mild steel of the gage specified for the operation being performed. These operations shall be performed without leaving burrs, wire edges, cracks, or other defects on the workpiece.

3.7 Standard equipment. Unless otherwise specified (see 6.2.1), the following standard equipment shall be furnished with each machine.

a. One set of tools normally furnished with each machine

b. One set each of the following rolls complete with any spacing collars, gages, or special nuts incidental to the rolls:

Turning rolls
Burring rolls
Wiring rolls
Flanging rolls
Crimping rolls
Ogee bead rolls
Single bead rolls
Slitting rolls

3.8 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2.1).

3.9 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The chart or plate shall contain the following information:

Points of lubricant application
Service interval
Type of lubricant
Viscosity

3.10 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion-resistant metal nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix indentified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:04Z
Check the source to verify that this is the current version before use.

Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture
U S

3.11 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.12 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. When first article approval is required under 3.1, first article inspection shall be performed. Unless otherwise specified (see 6.2.1), first article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure of the item to pass any examination or test shall be cause for disapproval of the first article.

4.2.2 Quality conformance inspection. Each item shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.3, the test in 4.4 through 4.4.2 and the inspection 4.5. Failure of the item to pass any examination, test, or inspection shall be cause for rejection.

4.3 Examination. The machine and equipment shall be examined to determine compliance with the requirements in 3.2 through 3.12.

4.4 Tests.

4.4.1 Operational test. The machine shall be operated at no load for not less than 15 minutes, during which time the controls and all operating components shall be actuated and observed to determine that they operate properly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:04Z
Check the source to verify that this is the current version before use.

4.4.2 Performance test. The machine shall be used to produce two workpieces, each having not less than 24 inches of formed edge, with each set of rolls furnished with the machine. The workpieces shall be of the gage specified in table I. The finished workpieces shall be free of burrs, cracks, or other defects.

4.5 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packing and marking. Unless otherwise specified (see 6.2.1), preservation, packing and marking shall conform to the requirements of MIL-STD-1188.

6. NOTES

6.1 Intended use. Machines covered by this specification are intended for use in sheet metal fabricating shops to perform wiring, turning, slitting, burring, crimping, and flanging operations.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.

b. Type and size machine required (see 1.2 and table I.).

c. First article approval, if required (see 3.1 and 6.3).

d. Specify additional safety and health requirements (see 3.2.3 and 6.4).

e. Painting, if different (see 3.3.4).

f. Electrical system requirements, if different (see 3.4.6).

g. Type motor enclosure, if different (see 3.4.6.1).

h. Standard equipment, if different (see 3.7).

i. Optional equipment, if required; fully describe (see 3.8).

j. Lubrication plate, if required (see 3.9).

k. Nameplate, if different (see 3.10).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:04Z
Check the source to verify that this is the current version before use.

l. First article inspection, if different (see 4.2.1).

m. Quality conformance inspection, if different (see 4.2.2).

n. Preservation, packing and marking, if different (see 5.1).

6.2.2 Contract data requirements. Required technical data, such as operators manuals, parts lists, wiring diagrams, and other instruction for operation and maintenance as identified on a numbered DD Form 1664 should be specified on a DD Form 1423 incorporated into the contract.

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The first article should be a preproduction sample, or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, tests and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.3 requires compliance with OSHA that concern the machine. Noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc. as outlined, due to the proposed operating environment, including additional point of operation guarding, as specified in 3.2.3 by the using agency should be furnished, as specified in detail (see 6.2.1(d).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review Activities:
Army - AL
Navy - SH
DLA - GS

User Activity:
Navy - YD, SH

Preparing Activity:
DLA - IP

Project Number:
3441-0101

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:04Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-F-3967D Forming Mchs, Sheet Metal, Rotary, Combination, Floor Mounting.

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:04Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE LOGISTICS AGENCY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 4966 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Industrial Plant Equipment Center
ATTN: DIPEC-TF
Memphis, Tennessee 38114

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:04Z
Check the source to verify that this is the current version before use.